Giant Hepatic Mesenchymal Hamartoma in an Infant: A Case Report of a Rare Tumour

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ABSTRACT
Primary paediatric liver tumours are very rare, accounting for 5-6 % of all intra-abdominal tumours. Mesenchymal hamartoma is the second most common benign liver tumour in children, with 80 % of the cases diagnosed in children less than 2 years of age. The aetiology is still unclear, but it is believed to be a developmental disorder rather than a true neoplasm. The presentation of these tumours is variable, ranging from small asymptomatic lesions to very large tumours filling the entire abdominal cavity and causing life threatening complications. Complete surgical excision is the gold standard of treatment for these lesions. We present a giant hepatic mesenchymal hamartoma in a 17-month-old female and discuss the presentation, pathology, diagnosis and therapeutic approaches for this condition.

Keywords: *Hepatic, Mesenchymal, hamartoma, liver tumour, Paediatric surgery*

INTRODUCTION

Primary paediatric liver tumours are rare, accounting for 5-6% of all intra-abdominal tumours.[1] Hepatic mesenchymal hamartoma (HMH) is a rare tumour of infancy and childhood, accounting for approximately 8 % of all tumours in the paediatric age group.[2] It is the second most common benign liver tumour in children after infantile haemangioendothelioma.[3] Although this pathology has been reported globally, fewer than 200 documented cases exist.[4] Eighty percent (80 %) of cases occur in children under 2 years of age, with 20 % identified during the neonatal period.[5] Clinical presentation varies, ranging from asymptomatic lesions, detected incidentally on imaging, to rapidly expanding cystic masses. Symptoms may include abdominal distention, respiratory distress, ascites, jaundice, and cardiac failure.[6] Several options for the management of these tumours have been described, including watchful waiting, enucleation, and marsupialization.[6] If neglected, the tumour may grow to compress adjacent structures, become unresectable or undergo malignant transformation. Therefore, early clinical detection and complete surgical excision are key. We report a huge mesenchymal hamartoma of the liver in a 17-month-old girl managed at a University Teaching Hospital, South-South of Nigeria.

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CASE REPORT

A 17-month-old female presented with a 2month history of abdominal distension and mass. There was no history of pain, vomiting, constipation or jaundice. On physical examination, she was chronically ill looking, wasted, pale with lower extremity oedema. Her temperature was normal, respiratory rate was 68/min, pulse rate was 124 beats per minute and her blood pressure was within normal range. She had abdominal distension with distended abdominal veins and a 28 x 28 cm mass arising from the right upper quadrant and filling most of the abdominal cavity (Figure 1).

Figure 1 - shows the giant mass filling most of the abdominal cavity.

The mass was not tender, smooth, and firm with well-defined edges. Other systems examined were normal. Laboratory findings showed a minimal elevation of transaminase levels and acid phosphatase. Her complete blood count, renal function test, and serum alpha-fetoprotein levels were normal. Abdominal ultrasound scan revealed an enlarged liver with an avascular, round hypodense, hypoechoic mass. The mass was contained in the right hepatic lobe and had cystic and solid components separated by thin septa. There was no evidence of portal hypertension, biliary obstruction or ascites, and there was no cystic involvement of the kidneys or any other organs. At exploratory laparotomy, the mass was on the anterior and inferior surface of the liver involving segments IV, V and VIII (Figure 2(a)).

Complete excision of the mass was done including a perimeter of normal liver tissue (Figure 2(b)). Her post-operative period was uneventful, and she was discharged on the 7^{th} day post-op. Microscopic examination of the excised mass (Figures 3) revealed a fibrocollagenous stroma containing variable sized lymphatic channels, blood vessels and ductular structures lined by cuboidal epithelium. The lymphatic channels contain eosinophilic to mucin like materials, and blood vessels were filled with red blood cells. No evidence of mitotic activity, atypia or malignant transformation was observed. The histological diagnosis confirmed mesenchymal hamartoma of the liver with negative surgical margins (Figures 3). The child was lost to follow-up after 6 months.

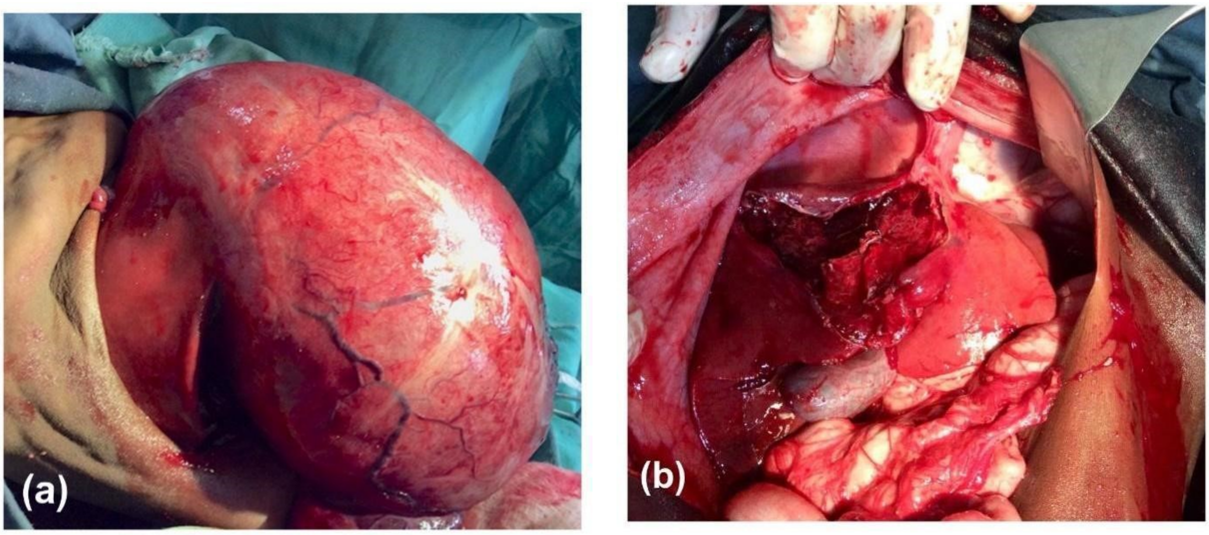

Figure 2 - Intraoperative findings. Panel (a), Tumour occupying segments IV, V and VIII. Panel (b), resection bed and remaining liver segments.

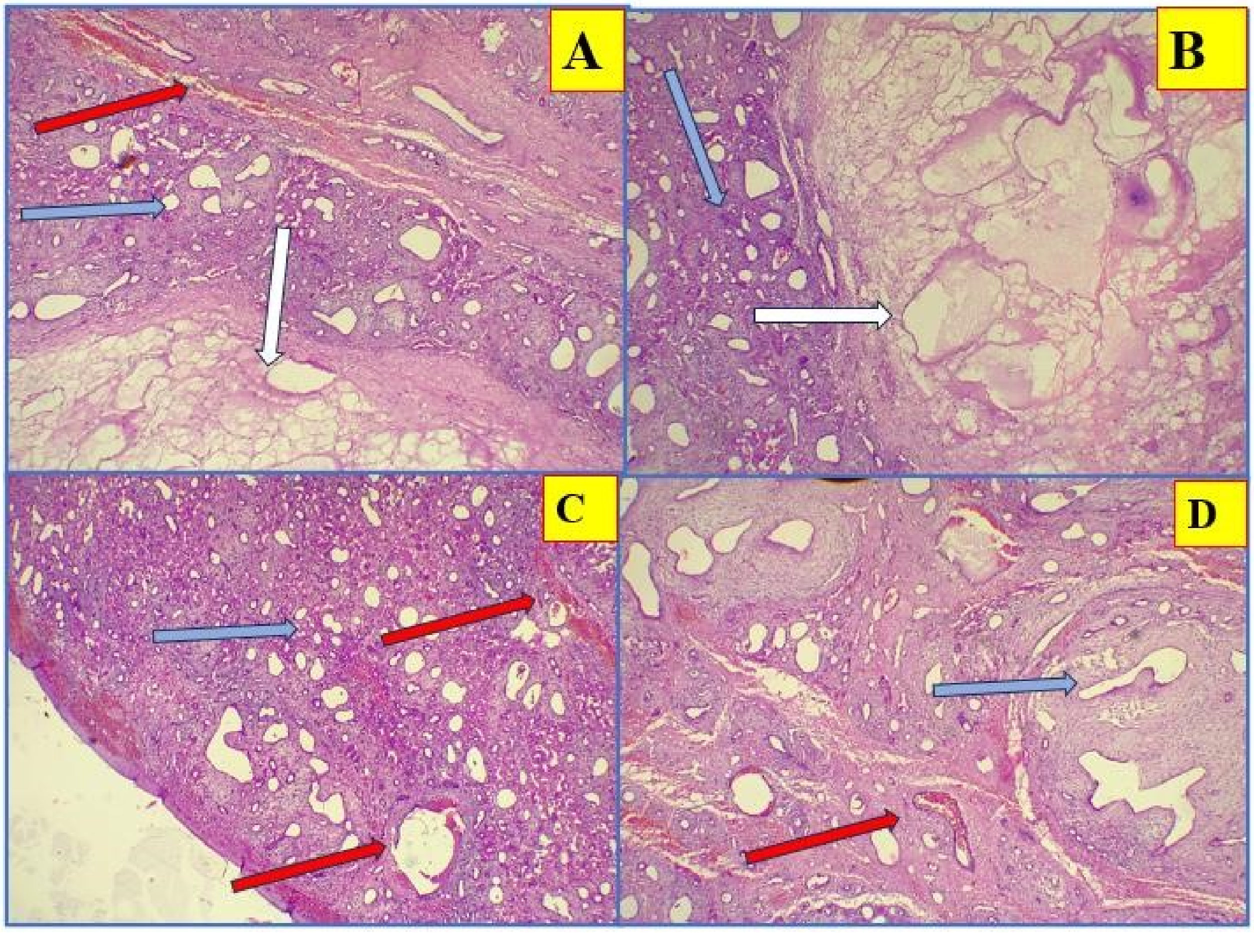

Figure 3 (A-D): Microscopic morphological features of the hepatic cystic mass showing admixed variably-sized lymphatic channels (white arrow), blood vessels (red arrow) and ductular structures (blue arrow) within the fibrocollagenous stroma (H&E x4).

DISCUSSION

Hepatic mesenchymal hamartoma (HMH) is a rare developmental tumour of the liver occurring during infancy and early childhood. The first documented description of HMH was by Edmondson in 1959, in his report of four patients.[7] HMH accounts for 8–10% of all paediatric tumours and is the second most common benign liver tumour in childhood, following infantile haemangioendothelioma.[3] As with our case, it is frequently diagnosed in children under the age of two, but it has also been discovered prenatally and in adults.[8-12] The median age of presentation is 10 months (0 – 19 years), with a male-to-female ratio of 2:1. There is no documented racial preference.[3,13-16]

The aetiology of HMH is unclear. Packard & Palmer and Denher et al. described HMH as a developmental anomaly of the foetal portal connective tissue.[13,17] Based on their theory, HMH is thought to be a malformation of the ductal plate where rests of mesenchymal tissue differentiate independently from the portal triad. This abnormal development of the ductal plate is believed to explain the similarities between the biliary duct abnormalities in this tumour and those in Caroli's disease.[18] Okeda and Von

Schweinitz et al cited in Stringer and Alazai, suggested a vascular insult or toxic injury to the fetal liver, as initiating factors for this lesion.[18-20] Another theory proposed that HMH is a true neoplasm, supported by recent cytogenic studies showing translocations involving chromosomes 11, 13, 15, and 19. Furthermore, the detection of aneuploidy in flow cytometry suggests a probable association between HMH and undifferentiated embryonal sarcoma. This means that HMH could be a precursor to cancer.[18] In contrast to simple non-parasitic cysts, HMH are large multicystic hepatic lesions. They arise from normal hepatic parenchyma and contain clear to yellow fluid or a gelatinous substance. The cysts range from a few millimeters to several centimeters in size. Cysts as large as 20-30 cm have been reported. These tumours lack a true capsule but are well circumscribed.[18] Seventy-five percent (75%) originate from the right lobe of the liver. Up to 20% of mesenchymal harmartomas are pedunculated, arising from the inferior surface of the liver.[16] Histologic findings showed solid areas with loose myxoid stroma and dilated ducts lined by cuboidal epithelium. Mixed within these solid areas are multiloculated cysts lacking an identifiable epithelial lining and appearing to be surrounded by mesenchyme.[18]

The typical symptoms include rapid but painless abdominal distention accompanied by a palpable abdominal mass on physical examination. The tumours rapid expansion is thought to be caused by mesenchymal degradation and fluid accumulation within the cysts.[16] Other presenting features caused by mechanical compression of the adjacent viscera may include failure to thrive, nausea, vomiting, jaundice, ascites, respiratory difficulty, and features of inferior vena cava compression. Our patient presented with abdominal distension, tachypnoea (due to diaphragmatic splinting by the giant tumour filling almost the entire abdominal cavity), lower limb oedema, and distended abdominal veins (in keeping with inferior vena cava compression).

HMH is not typically associated with congenital abnormalities, However, various abnormalities have been reported with HMH. These include mesenchymal hyperplasia of the placental stem villi, hydrops fetalis, congenital adrenal cytomegaly, small bowel malrotation, neonatal hyperbilirubinemia, biliary atresia, congenital heart disease, oesophageal atresia, annular pancreas, exomphalos, beckwith-Wiedemann syndrome, and diffuse endocrinopathy.[6,14,18,21] In addition, haematologic disorders such as idiopathic thrombocytopenic purpura and disseminated intravascular coagulation, also known as Kasabach-Merritt phenomenon, have been reported in older children.[6,22]

There are no tumour markers or laboratory investigations specific for HMH. Alpha fetoprotein is usually within reference range or mildly elevated and thus this test is used to exclude diseases like hepatoblastoma. Liver function tests are usually normal, although liver enzymes are occasionally elevated. Imaging studies are the most helpful modalities in the diagnosis of HMH which appears on Ultrasound scan and CT scan as large multi-cystic mass containing areas of low attenuation with internal septations which are enhanced with intravenous contrast administration.[23] Ultrasound is useful in confirming the hepatic origin of the mass, while a CT scan allows for the anatomical definition of the mass, which is important for surgical planning. In our setting, we frequently rely on ultrasound scans as the only imaging modality. This is because it is readily available, comparatively inexpensive and useful in resource-poor settings.[24] Our patient had an abdominal ultrasound scan, which confirmed the large liver mass with cystic and solid components.

In patients with asymptomatic HMH, the typical progression involves rapid enlargement followed by a decrease in size, prompting some authors to recommend a watchful waiting.[18] However, other authors discouraged watching the tumour for spontaneous regression because of its potential for malignant transformationto undifferentiated embryonal sarcoma (malignant mesenchymoma).[3,18,25] The recommended treatment comprises complete excision of the mass with clear margins. A recent systematic review by Bauzon et al.[26] concluded that complete tumour resection had a low morbidity rate and should remain the gold standard. This can be achieved by a formal lobectomy or resecting the hamartoma with a surrounding rim of normal liver parenchyma. Marsupialization or enucleation may be necessary when the tumour is deemed unresectable.[27] Other options for an unresectable tumour include Living Donor Liver transplant or Orthotopic Liver transplant.[4] However, these procedures are not readily accessible in our region. In our patient, we felt the need for prompt surgical intervention. This was due to the substantial size of the tumour, which showed features of splinting the diaphragm. The decision to excise the tumour as an emergency not only prevented the danger of a life-threatening respiratory distress but also offered a definitive treatment for the mesenchymal hamartomas.

Our patient had an uneventful post-operative course and follow up. This is similar to other reports where patients recovered with good outcomes. However, post-operative complications affecting up to 30-35% of patients have been reported,[4] including major morbidities such as post-operative infections, bile leaks, and incisional hernias. These complications are amenable to treatment with no long-term consequences.

CONCLUSION

Hepatic Mesenchymal hamartoma is considered a rare, benign, congenital childhood tumour. The aetiology is yet to be fully understood. Due to its diverse clinical presentation, it can mimic a variety of benign and malignant tumours. Typically, HMH presents as a large multicystic liver mass in a child younger than 2 years and must be considered as a possible diagnosis in children with huge abdominal masses related to the liver. Emergency intervention is often required, and the preferred treatment is a complete surgical excision to alleviate pressure symptoms and prevent future complications.

ETHICAL CONSIDERATIONS

The case report was conducted in compliance with the guidelines of the Helsinki declaration on research in human subjects and we obtained informed consent from the patient's mother for their anonymized clinical information and images to be published.

CONFLICT OF INTEREST

We declare no conflict of interest.

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